

Steven (Shang) Shi

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[Website](#) and [LinkedIn](#)

Education

Carnegie Mellon University

Master of Science in Mechanical Engineering (MS) – GPA: 3.97/4.0

Focus: **Machine Learning** and **Controls**

Relevant Courses: 24774-Advanced Control Systems, 16745-Optimal Controls, 16824-Visual Learning and Recognition, 11785-Deep Learning

Pittsburgh, PA

December 2023

Skills

Simulation & Modelling: SolidWorks, AutoCAD, Ansys Mechanical, RoboDK, AUBO Studio

Programming Languages: Python, C++, MATLAB, Julia, LUA

Tools/Platforms: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, LaTeX, Git, AWS, GCP, CRM, OpenAI SDKs

Work Experience

Robotics Business Development – AUBO Robotics USA (Romulus, MI)

Feb 2024 – Present

- Drive business development and partner acquisition, signing distributors and integrators while shaping pricing strategies to improve sales by over 2x and expanding market share.
- Serve as technical-commercial lead, training partners and aligning stakeholders to drive successful deployments.
- Develop and integrate cobot solutions (case packing, palletizing, vision-guided pick-and-place) with PLCs, sensors, and custom EOAT.
- Identify top-performing lead sources and build a marketing intern team using AI-driven tools to scale outreach and increase qualified pipeline.

Mechanical Engineering Intern – AGS Automotive Systems (Toronto, ON)

Jun 2020 – Oct 2021

- Managed 10+ projects for a total spend of \$200k USD
- Negotiated vendor contracts, saving ~\$160k USD through 3-years.
- Coordinated cross-functional communication between plant teams, vendors, and senior management.
- Presented technical and business recommendations to executives, including the company president.

Engineering Design Projects

Dense Video Captioning with Semantic Alignment – Deep Learning

Spring 2023

- Led team in enhancing a state-of-the-art dense video captioning pipeline and surpassing baseline results.
- Trained deep learning models using Pytorch for semantic alignment between visual and caption features.
- Integrated semantic alignment models with the captioning pipeline, achieving an **8% improvement** across all metrics compared to baseline

Control Stabilization for Dual-Quadrotor Payload Delivery

Fall 2023

- Led team to modify embedded firmware (C) for dual-drone payload transport.
- Developed Python API scripts for synchronized swarm control.
- Built velocity vs. set point plotting and error analysis pipeline for rapid controller tuning.
- Tuned and tested Cascaded PID controller based on MATLAB simulation; designed and implemented payload-based gain adjustments
- Achieved **+43% payload capacity** and **-26% error percentage**.

Diffusion Based Comic Generation Pipeline with Few-shot Finetuning

Fall 2023

- Led team to design text-image-image diffusion pipeline for generating coherent comic panels from user prompts.
- Implemented ControlNet with canny edge maps to preserve content during style transfer.
- Applied a feature-based style loss metric to automatically select the best stylistic match from generated batch.
- Fine-tuned few-shot style transfer model by adjusting guidance scale and inference steps leading to noticeable qualitative improvements.